Principal component methods

Readings for today

- Chapter 6: Linear model selection and regularization. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: with applications in R (Vol. 6). New York: Springer

Topics

1. Principal components analysis

2.Factor Analysis

3. Principal component regression

4. Partial least squares

Principal component analysis

Dimensionality

<u>Dimensionality of a model</u>: n x p

As $n \rightarrow p$, dimensionality increases & model variance increases

$$\begin{pmatrix} x_{1,1} \\ \cdots \\ x_{n,1} \end{pmatrix} \rightarrow \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,15} \\ \cdots & & & \\ x_{n,1} & x_{n,2} & \cdots & x_{n,15} \end{pmatrix} \rightarrow \uparrow \text{model flexibility}$$

How do we reduce the dimensionality of our model?

Can we do this in a way that preserved or compresses meaningful information in our feature set?

(James et al. 2013)

How to deal with high model dimensionality

So far we have covered:

1. Feature selection by comparing lower dimensional variants of your model.
   - Best subset selection
   - Forward/Backward stepwise selection
2. Apply a sparsity constraint to your model during fitting.
   - Ridge regression
   - LASSO
   - Elastic Net

(James et al. 2013)

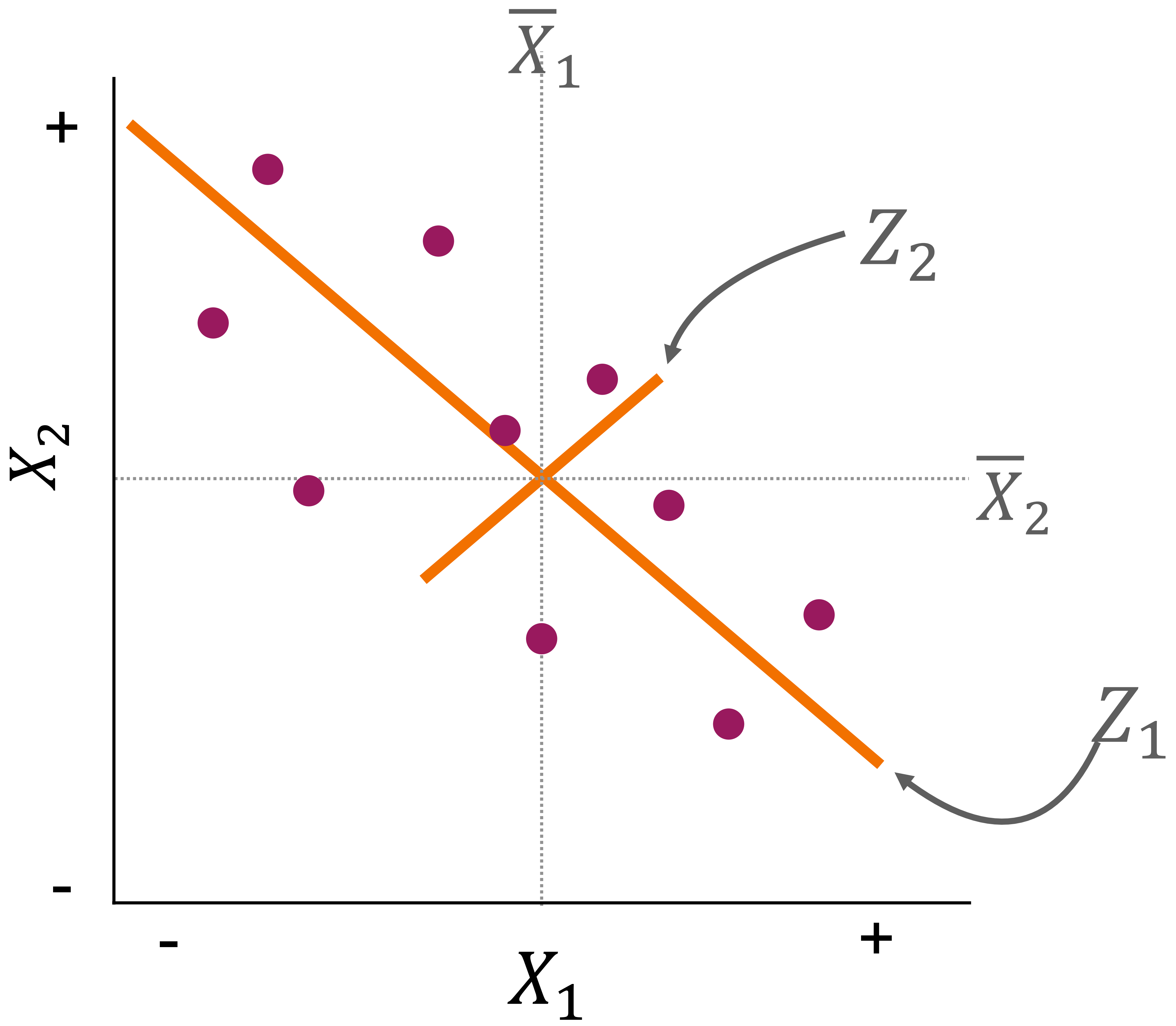
Principal component analysis (PCA)

What if you reduce the dimension of X itself?

<u>PCA:</u>

$$\underbrace{Z_m}_{\text{lower dimensional projection (component)}} = \sum_{j=1}^{p} \underbrace{\phi_{j,m}}_{\text{component loading}} \underbrace{X_j}_{\text{original data variable}}$$

Low dimensional components



The first principal component (Z_1) explains the most variance about the relationship between X_1 and X_2.

PCA algorithm

Step 1: Find the first component (Z_1) loading.

$$\phi_1 = \text{arg}max(\frac{\phi_1' X' X \phi_1}{\phi_1' \phi_1})$$

Step 2: Take the residuals after accounting for Z_1.

$$\hat{X}_m = X - \sum_{s=1}^{m-1} X \phi_s \phi_s'$$

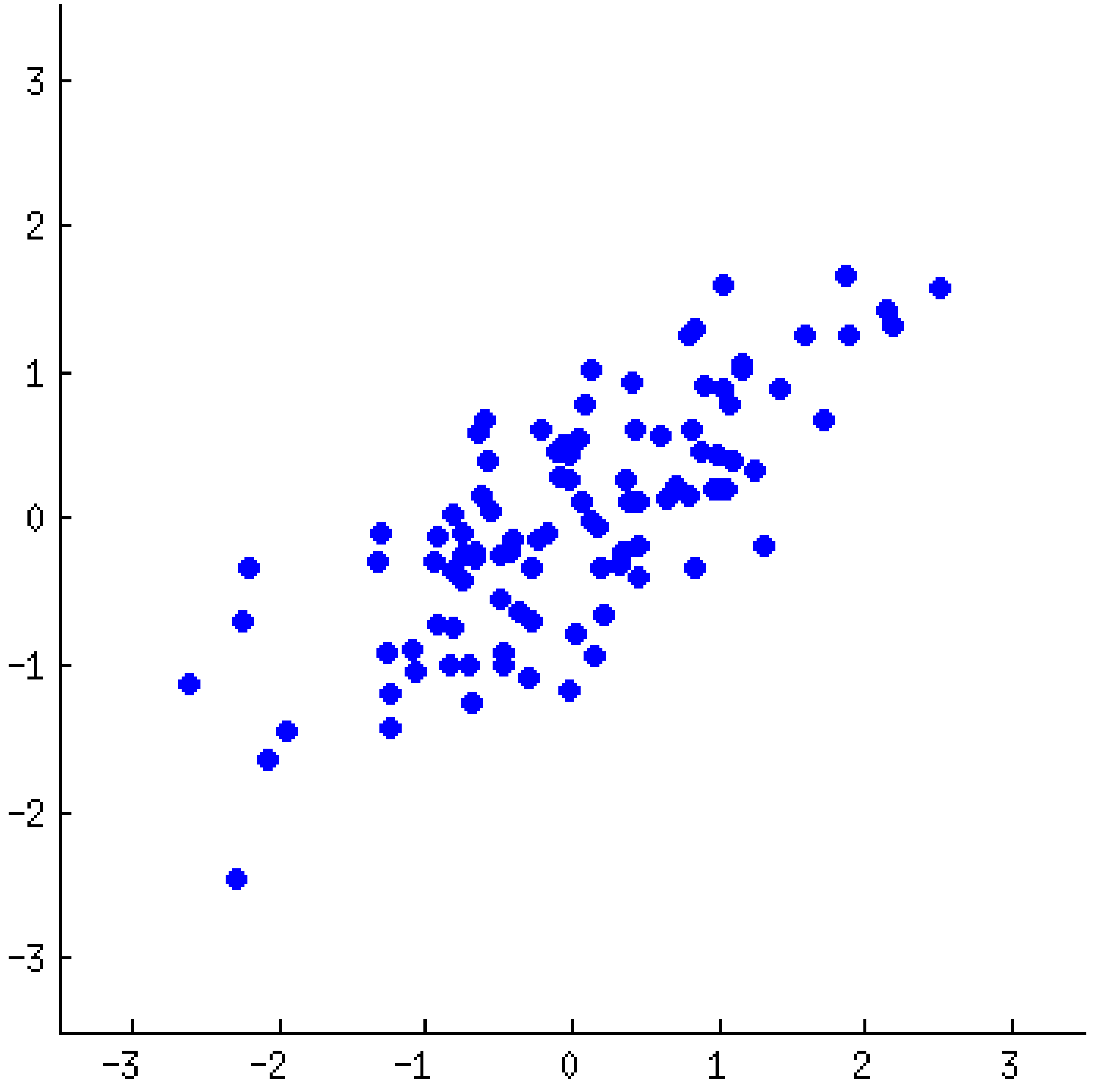
Step 3: Calculate the next component (Z_m).

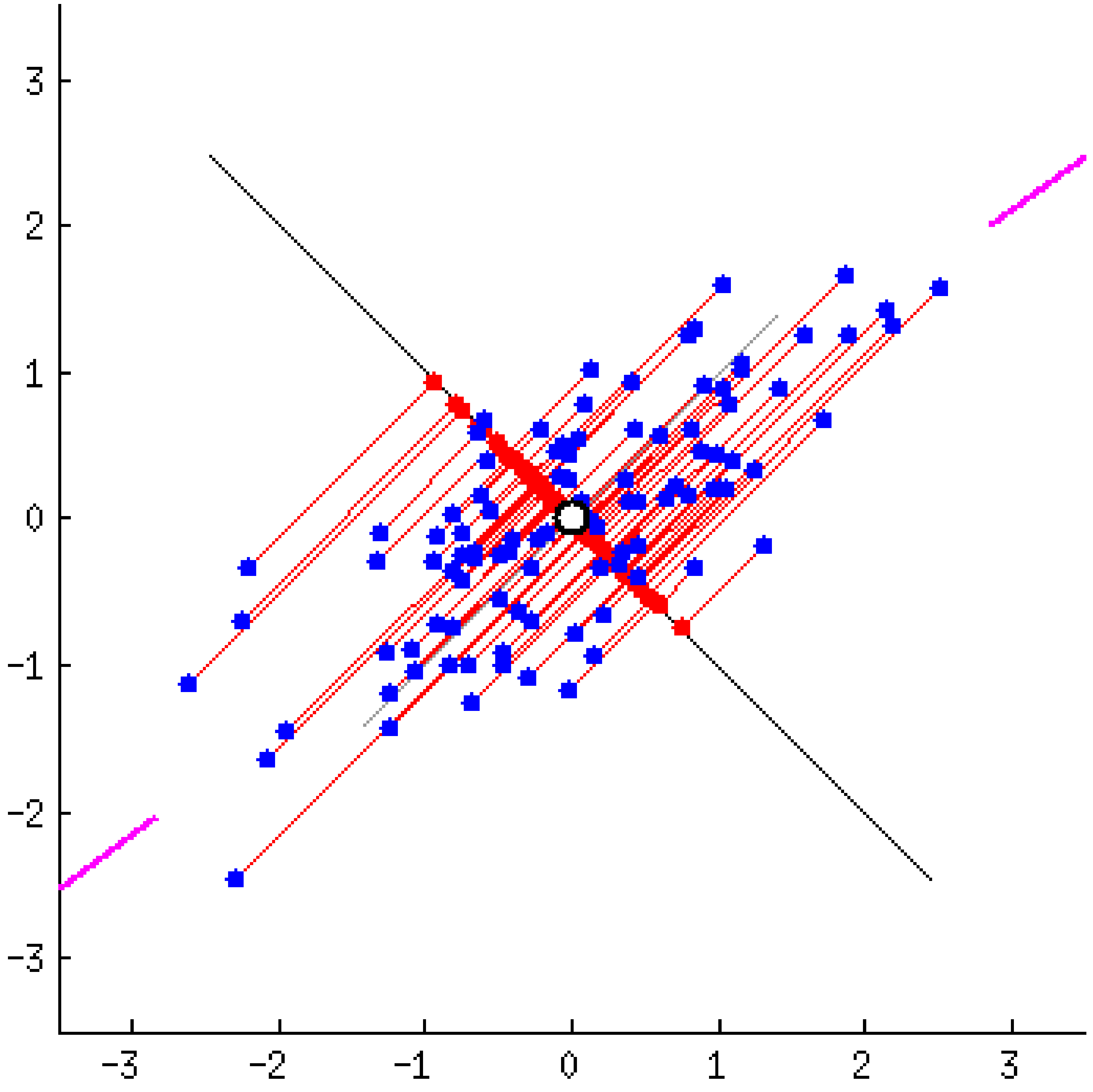
$$\phi_m = \text{arg}max(\frac{\phi_m' \hat{X}_m' \hat{X}_m \phi_m}{\phi_m' \phi_m})$$

Step 4: Repeat Steps 2-3 until $m = p$

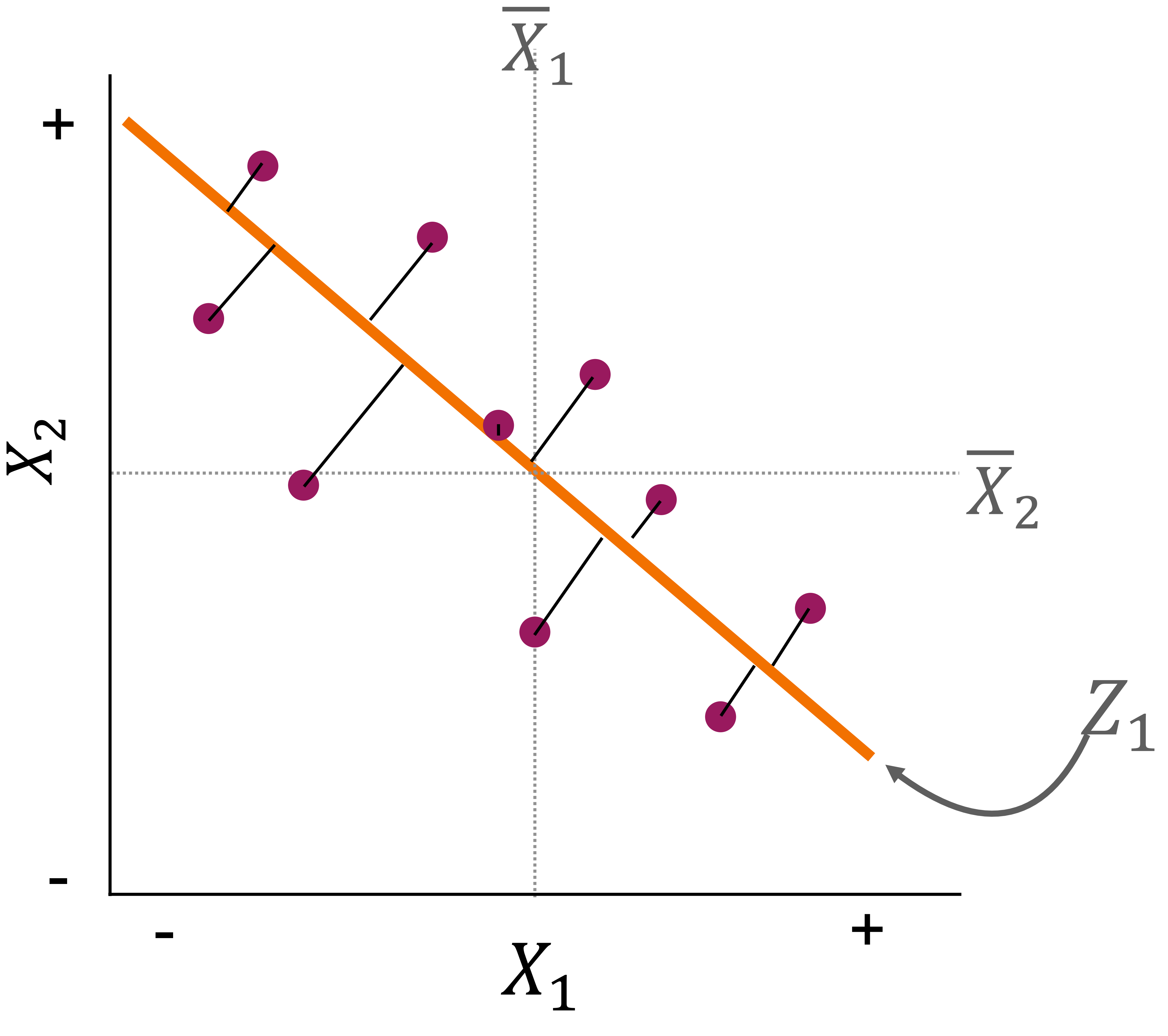
(James et al. 2013)

PCA algorithm



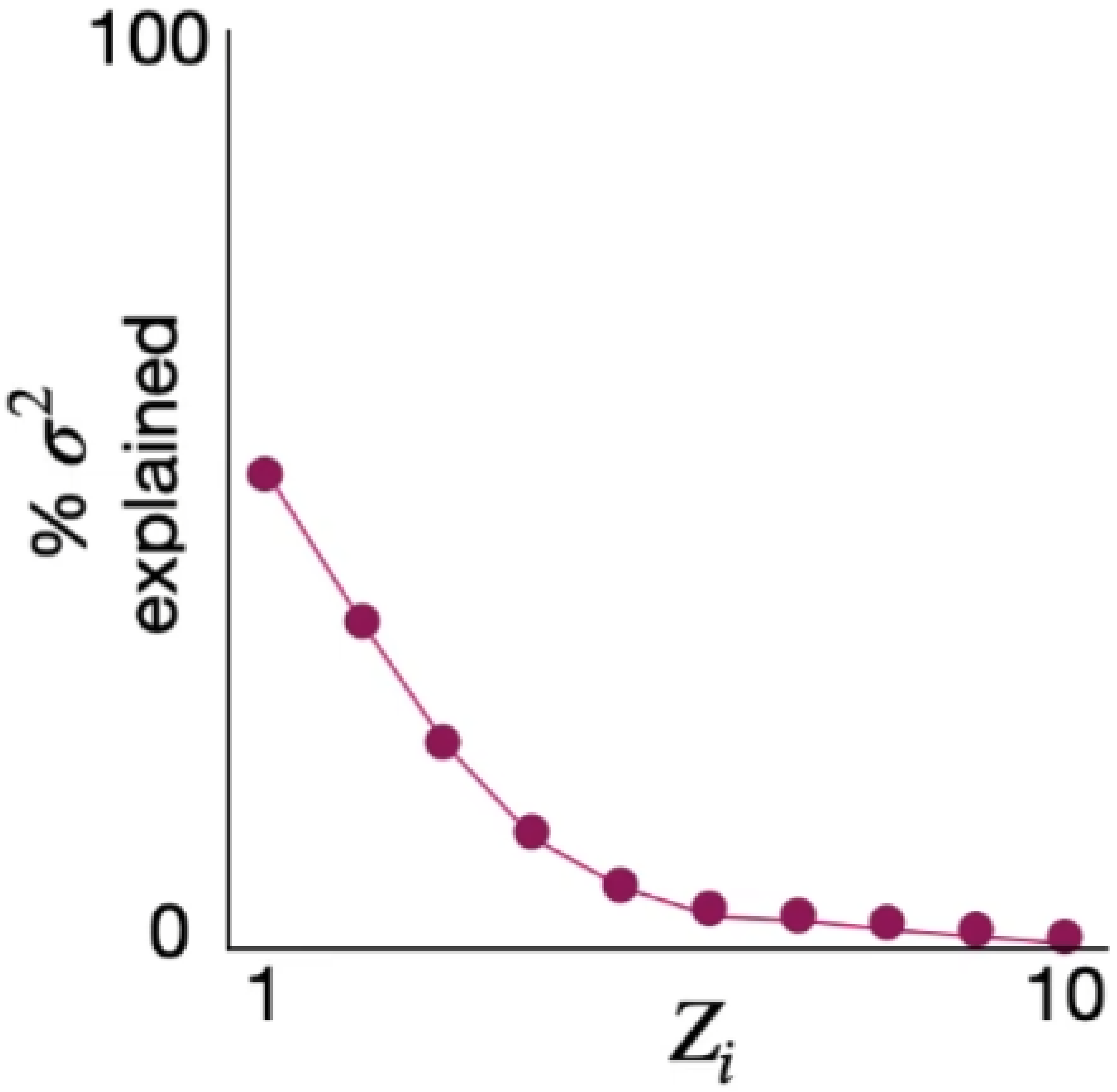


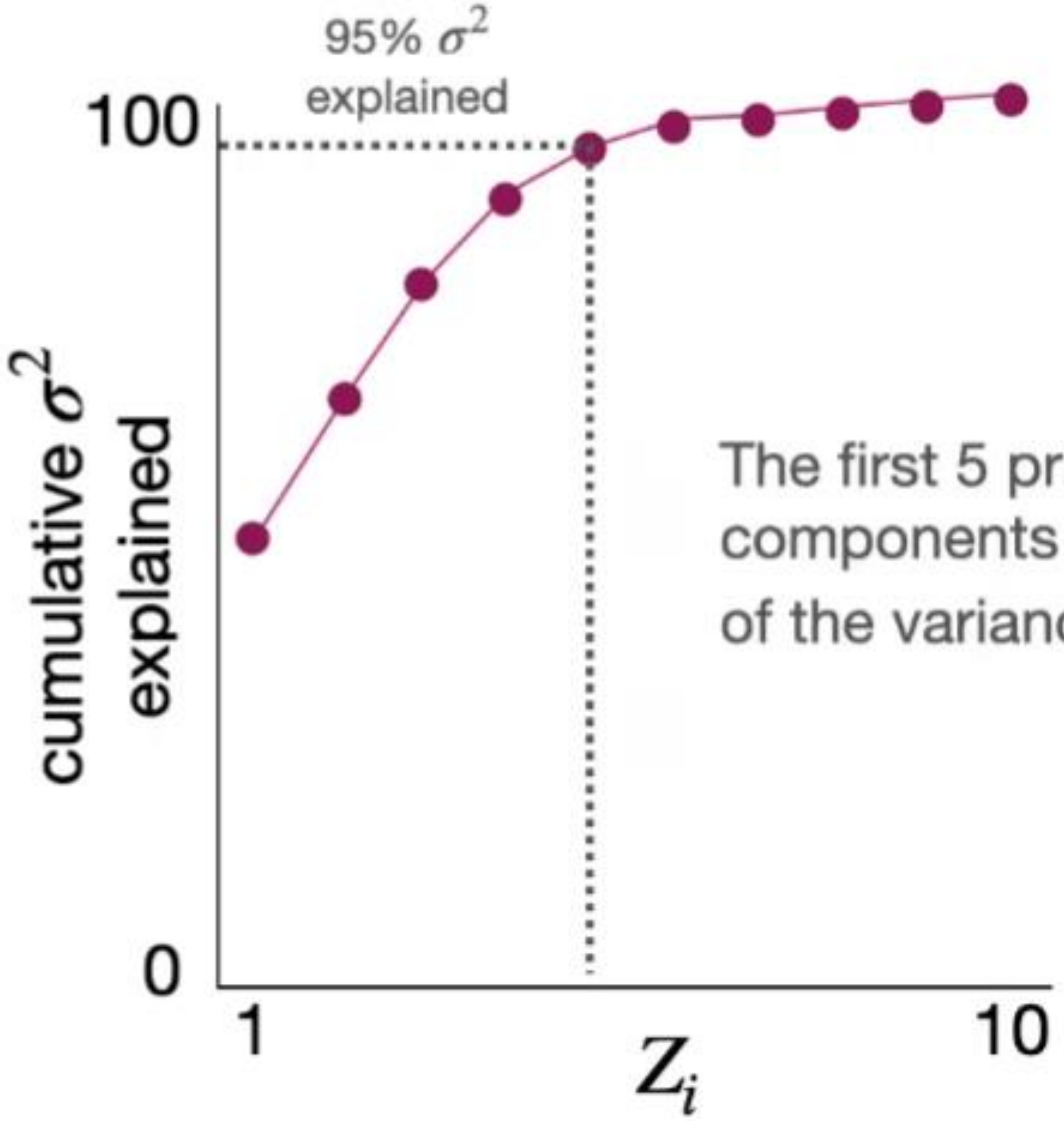
PCA algorithm



Example: 10-dimensional X

$$\begin{pmatrix} x_{1,1} & \dots & x_{1,p} \\ \vdots & & \\ x_{n,1} & \dots & x_{n,p} \end{pmatrix} \rightarrow n = 15,\ p = 10$$





The first 5 principal components explain 95% of the variance in X.

Factor Analysis

Factor Analysis

<u>Factor Analysis:</u> $\Sigma \approx \Lambda\Lambda' + \Psi$

$$X^T = L \quad F$$

$$\underbrace{}_{p\times n} \qquad \underbrace{}_{p\times k}\ \underbrace{}_{k\times n}$$

Factor loading
matrix with
k specified up front.

Λ: $p \times k$(these are your directions/’components’)
Ψ: diagonal noise

Iteratively tries to solve to
minimize difference between Σ and $\Lambda\Lambda'+\Psi$

Researcher specifies k

PCA vs. Factor Analysis

PCA: $$\underbrace{X}_{n\times p} = \underbrace{Z}_{n\times m}\underbrace{\phi^{-1}}_{m\times p}$$

Factor Analysis (FA): $$\underbrace{X^T}_{p\times n} = \underbrace{L}_{p\times m}\underbrace{F}_{m\times n}$$

Factor loading matrix with m specified up front.

Are you trying to *summarize* the data, or *explain* it?

- PCA is better for summarizing or exploring your data.
- FA is better for hypothesis testing & discovering latent constructs
- PCA explains all variance in X.
- FA assumes lower dimensionality in X.

Component/Factor Scores and Loadings

<u>Component/Factor Scores:</u>

$n \text{ x } Z$

$n \text{ x } F$

Each element is the degree that each <u>**observation**</u> aligns with the component or factor

<u>Component/Factor Components:</u>

$p \text{ x } Z$

$p \text{ x } F$

Each element is the degree that each <u>**feature**</u> aligns with the component or factor

Principal component regression

Principal component regression (PCR)

Goal: Reduce the dimensions of X using PCA and use the most explanatory components in Z (component **scores**) as your predictor variables.

PCA + linear regression:

$k < p$

# of components that explain $1 - \alpha$ percent variance in X

regression coefficients learned on $Z_1, \ldots, Z_k$

$$\hat{y}_i = \sum_{m=1}^{k} \hat{\theta}_m Z_{i,m}$$

(James et al. 2013)

PCR algorithm

Step 1: Calculate $Z = \phi X$.

Step 2: Identify the k components that explain $1 - \alpha$ (e.g., $(1 - 0.05) = 95\%$) of the variance in X.

Step 3: Fit your regression model with the k components identified in Step 2 (i.e., learn $\hat{y} = Z_k \theta + \epsilon$)

(James et al. 2013)

Projecting back to X

Can determine the regression weights in X (i.e., $\hat{\beta}_j$) that best resolve the bias-variance tradeoff via the coefficients learning in Z (i.e., $\hat{\theta}_j$)

PCR to linear regression:

$$\hat{y}_i = \sum_{m=1}^{k} \hat{\theta}_m Z_{i,m} \longrightarrow Z_m = \sum_{j=1}^{p} \phi_{j,m} X_j$$

$$= \sum_{m=1}^{k} \hat{\theta}_m \sum_{j=1}^{p} \phi_{j,m} X_j$$

$$= \sum_{j=1}^{p} \underbrace{\sum_{m=1}^{k} \hat{\theta}_m \phi_{j,m}}_{\hat{\beta}_j} X_j$$

$$\beta = \phi_k \hat{\theta}$$

Best regression model:

$$\hat{\beta}_j = \sum_{m=1}^{k} \hat{\theta}_m \phi_{j,m}$$

Even when n is close to p.

(James et al. 2013)

Partial least squares

Partial least squares (PLS)

<u>Goal:</u> Find the lower dimensions in X that *maximize* $Cov[X,Y]$

# of variables in Z ($k = p$)

# of variables in X

regression coefficients learned on $Z_1, \ldots, Z_k$

$$\hat{y}_i = \sum_{j=1}^{p} \sum_{m=1}^{k} \hat{\theta}_m \hat{\phi}_{j,m} x_{i,j}$$

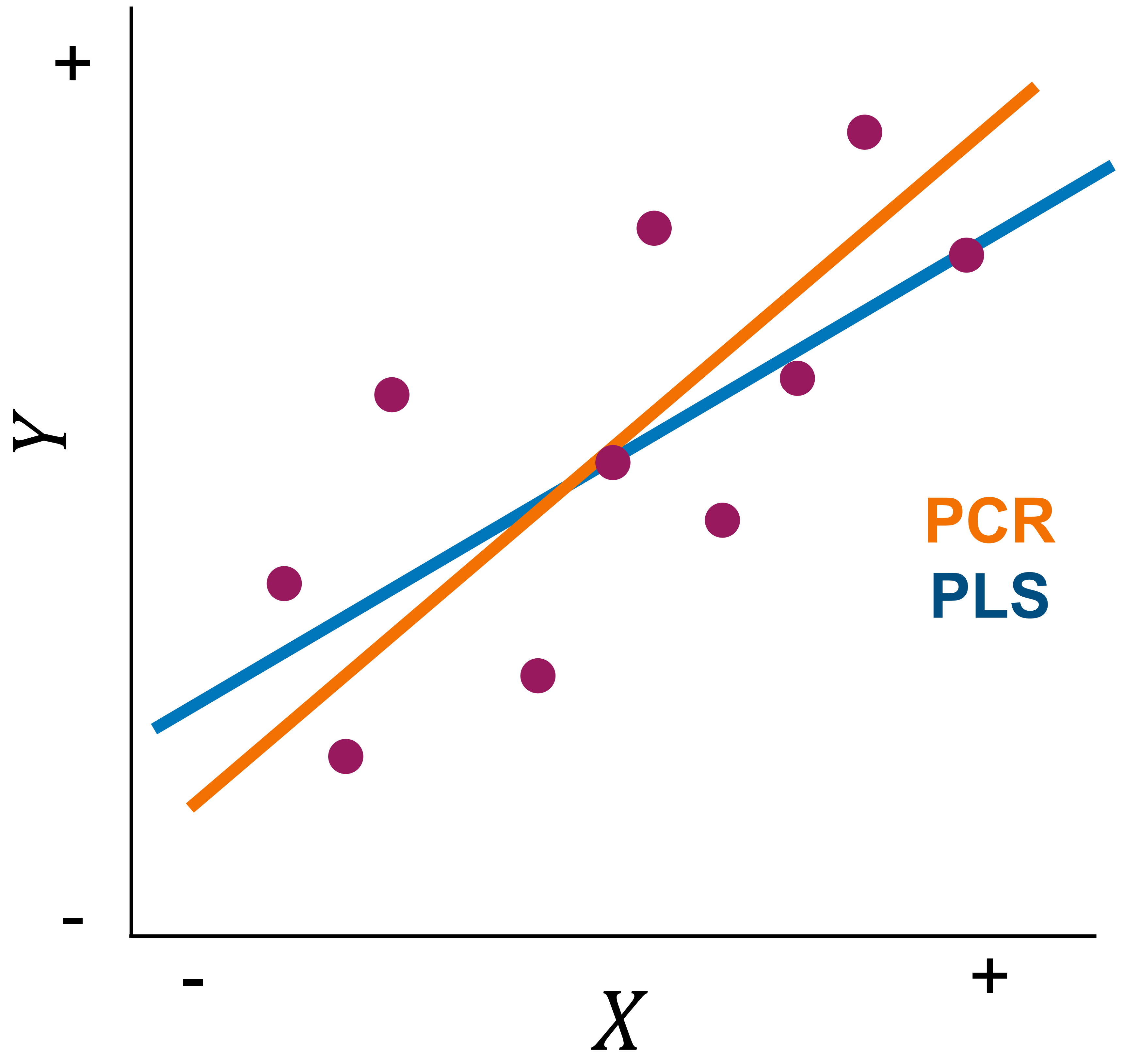
principal component loading learned by minimizing

$$\sum (y - \sum_{m=1}^{k} \hat{\theta}_m Z_{i,m})^2$$

$\hat{\phi}$ and $\hat{\theta}$ estimated <u>in one step.</u>

(James et al. 2013)

PLS vs. PCR



- PLS and PCR produce qualitatively different results depending on how the low dimensional components in X associate with Y.

(James et al. 2013)

Take home message

- Principal component methods offer an easy way of resolving the bias-variance tradeoff in high dimensionality (i.e., high variance) contexts by leveraging any correlational structure in your predictor variables.